Data Science for Political Campaigns

Week 9 — Cost-effectiveness across tactics

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Roadmap

- **Last week:** three polls of the same race disagreed by 12 points. Nonresponse bias, mode effects, and the limits of reweighting.
- **This week:** four GOTV tactics. Which should a donor fund?
- **Next week:** turnout models.

This Week's Case

Case: You are a research analyst advising a political donor

The donor has $500,000 to spend on GOTV in competitive state legislative races. She wants the most additional votes for her money.

Four tactics are on the table. How do you split the budget?

The Four Tactics

Tactic	Effect	Cost/contact	Source
Canvassing	+2.0 pp	$20	Green & Gerber
Direct mail	+0.7 pp	$0.60	Gerber et al. 2017
Phone calls	+0.5 pp	$2.00	Green & Gerber
Text messages	+0.1 pp	$0.08	recent SMS studies

Discuss: What should she fund?

(These are the donor's working numbers. They come from four separate studies in four different elections. Course readings will not match them.)

Cost Per Additional Vote

Cost per additional vote: what it costs to produce one more vote than you would have had.

Mail 1,000 people.

- Cost: $1{,}000 \times \$0.60 = \$600$
- Votes: $+0.7$ pp is 0.7 of every 100 people, so $1{,}000 \times 0.007 = 7$ additional votes
- Cost per vote: $\$600/7 \approx \$86$

The 1,000 cancels, so you can do it one person at a time:

$$\text{cost per vote} = \frac{\text{cost per person}}{\text{effect (as a proportion)}} = \frac{\$0.60}{0.007} \approx \$86$$

Watch the units! "As a proportion" means the percentage-point effect divided by 100: $+0.7$ pp $= 0.007$. Using 0.7 makes the vote look $100\times$ too cheap.

Knock on 1,000 Doors

Same count for canvassing: \$20 a conversation, +2.0 pp. Canvassers reach about 30% of the doors they knock (Week 5).

- Cost: 300 doors open, so $300 \times \$20 = \$6{,}000$
- Votes: $1{,}000 \times 0.02 = 20$ additional votes
- Cost per vote: $\$6{,}000/20 = \$300$

The shortcut gives $\$20/0.02 = \$1{,}000$. That is wrong. You pay \$20 only at the 300 doors that open. But +2.0 pp is averaged over all 1,000 doors, including the 700 where nobody answered.

Per Door or Per Conversation

This is Week 7. +2.0 pp is the ITT: the effect per door you knocked. The effect on the people you actually reached is the LATE: $2.0/0.30 = 6.7$ pp.

Either one works, as long as the cost matches:

	Cost	**Effect**	**Cost per vote**
Per door knocked	$\$20 \times 0.30 = \$6$	+2.0 pp (ITT)	$\$6/0.020 = \$300$
Per conversation	\$20	+6.7 pp (LATE)	$\$20/0.067 \approx \$300$
Mixed up	\$20 per conversation	+2.0 pp per door	\$1,000

Before you divide, check that the top and the bottom describe the same people.

Cost Per Vote, All Four

Tactic	Effect	Cost/attempt	Cost/vote
Text messages	$+0.1$ pp	\$0.08	\$80
Direct mail	$+0.7$ pp	\$0.60	\$86
Phone calls	$+0.5$ pp	$\$2.00 \times 0.25 = \$0.50$	\$100
Canvassing	$+2.0$ pp	$\$20 \times 0.30 = \$6.00$	\$300

Assume 25% of phone calls and 30% of door knocks result in a contact.

Canvassing has the biggest effect and the most expensive vote: \$300 against mail's \$86, about 3.5 times the price.

Net Votes

A campaign does not need votes. It needs more votes than the other side: **net votes**.

GOTV mail turns out 100 people who were not going to vote. Say they break 75–25 for your candidate:

$$75 - 25 = 50 \text{ net votes}$$

100 additional voters cost $100 \times \$86 = \$8{,}600$, so a net vote costs $\$8{,}600/50 = \$172$.

\$86 per additional voter, \$172 per net vote for your side. At 50–50 it buys no net votes at all, which is why targeting matters.

Persuasion Counts Twice

Persuasion changes the minds of people who were already voting. Each voter you switch leaves the other column and joins yours: -1 for them, $+1$ for you.

	Yours	**Theirs**	**Net votes**
Turn out 100 voters who break 75–25	$+75$	$+25$	$+50$
Persuade 25 of their voters	$+25$	-25	$+50$

25 persuaded voters are worth the same as 100 mobilized ones (when 75% of the mobilized voters support you).

Confidence Intervals and Costs

Every effect estimate has a confidence interval. Mail's is $[+0.3, +1.1]$. Price each end of it:

		Effect	**Cost per vote**
Top of the interval	(best case)	$+1.1$ pp	$\$0.60/0.011 = \$55$
Point estimate		$+0.7$ pp	$\$0.60/0.007 = \$86$
Bottom of the interval	(worst case)	$+0.3$ pp	$\$0.60/0.003 = \$200$

A *bigger* effect means a *cheaper* vote, so the top of the effect's interval is the bottom of the price's.

\$86 is not in the middle of \$55 and \$200. Dividing by a small number blows up, so the worst case sits much further away than the best case.

Uncertainty Changes the Ranking

Same arithmetic, all four tactics:

Tactic	Effect interval	Best case	Point estimate	Worst case
Texts	$[-0.1, +0.3]$	\$27	\$80	never
Mail	$[+0.3, +1.1]$	\$55	\$86	\$200
Phone	$[+0.1, +0.9]$	\$56	\$100	\$500
Canvassing	$[+1.0, +3.0]$	\$200	\$300	\$600

At the point estimates, texts are the cheapest vote. At the worst case, they are the most expensive: the interval includes zero, and at zero effect no amount of money buys a vote.

Mail's worst case, \$200, is the best worst case of the four.

Bayesian Updating

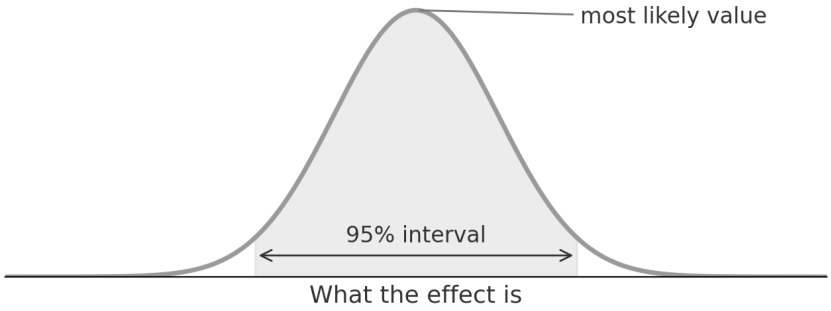
The older literature says canvassing moves turnout +2.0 pp. A new 2024 study says +0.8 pp. What should you believe now?

Bayesian updating: combining what you believed before with what new evidence says, to get what you believe now.

- **Prior:** what you believed before the new evidence arrived.
- **Updated belief:** what you believe after. (You will also see it called the *posterior*.)

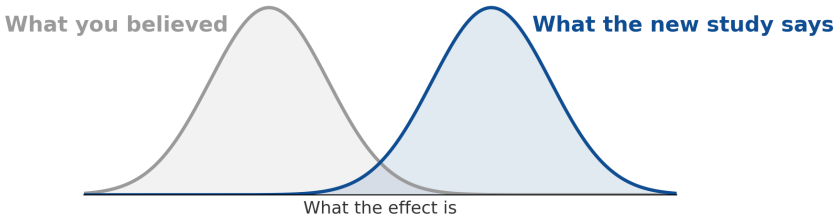
Let's start with pictures.

Picturing a Belief



Draw a belief as a curve. The peak is your best guess. The width is how sure you are: narrow means confident, wide means not. That width comes from the standard error.

Two Beliefs



Two curves: what you believed (grey) and what a new study says (blue). Here they are equally sure (similar standard errors), so the curves are the same width.

Where should you land after seeing the new evidence?

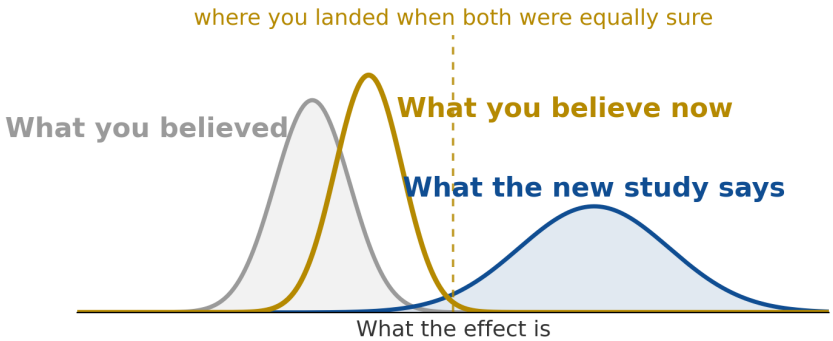
Equally Sure: Meet in the Middle



When the two are equally sure, you land halfway between them.

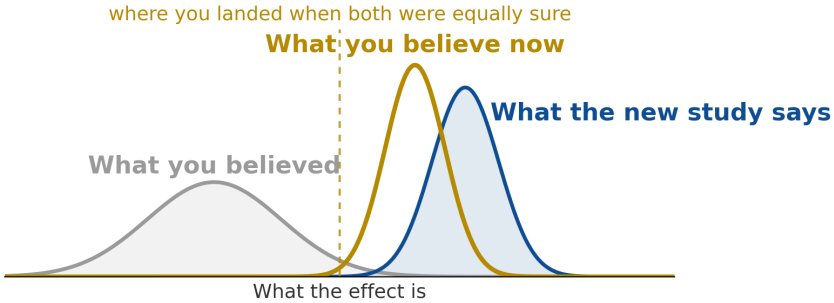
The updated curve is also narrower than either one. Two pieces of evidence make you more sure than one.

A Tighter Prior Moves Less



Same new study, but this time you were much more sure to begin with (half the SE). You move only a fifth of the way toward the study. A confident prior takes a lot of evidence to move.

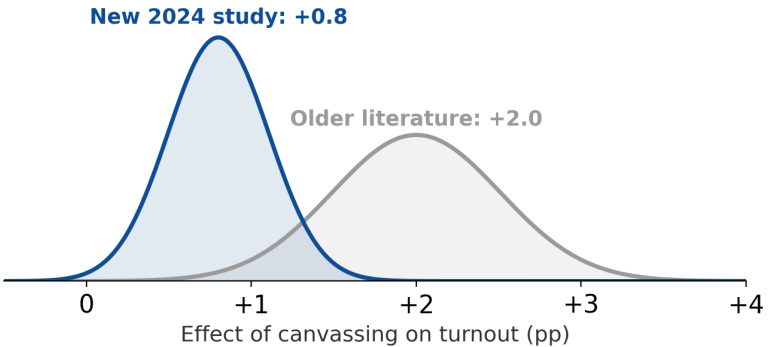
A Tighter Study Moves You More



Back to the original prior, but now the new study is much more precise. You move four fifths of the way.

You move toward whichever belief is narrower.

Back to Canvassing



Older literature: $+2.0$ pp, interval $[+1.0, +3.0]$, dozens of experiments in many settings.

New 2024 study: $+0.8$ pp, interval $[+0.2, +1.4]$, $n = 107{,}000$.

Precision

The updated belief is a weighted average of the two. Each one's weight is its **precision**:

$$\text{precision} = 1/\text{SE}^2$$

A small SE means a narrow curve, and a narrow curve gets a big weight (you are really confident in it).

Get each SE from its interval. A 95% interval is the estimate ± 1.96 SE, so SE = half the interval's width / 1.96.

	Prior	**New study**
Estimate	$+2.0$ pp	$+0.8$ pp
95% interval	$[+1.0, +3.0]$	$[+0.2, +1.4]$
Half the width	$(3.0 - 1.0)/2 = 1.0$	$(1.4 - 0.2)/2 = 0.6$
SE	$1.0/1.96 = 0.5102$	$0.6/1.96 = 0.3061$
Precision	$1/0.5102^2 = 3.84$	$1/0.3061^2 = 10.67$

Doing the Arithmetic

Step 1. Turn the precisions into weights (each one's share of the total, $3.84 + 10.67 = 14.51$).

$$\text{weight}_{\text{prior}} = \frac{3.84}{14.51} = 26\% \qquad \text{weight}_{\text{new}} = \frac{10.67}{14.51} = 74\%$$

Step 2. Take the weighted average.

$$0.26 \times 2.0 \;+\; 0.74 \times 0.8 \;=\; 0.52 + 0.59 \;=\; +1.1 \text{ pp}$$

The Updated SE

Precision measures how much information you have. Putting the two together gives you the information from both, so you add their precisions:

$$\text{updated precision} = 3.84 + 10.67 = 14.51$$

Turn that back into an SE. Precision is $1/\text{SE}^2$, so SE is $1/\sqrt{\text{precision}}$:

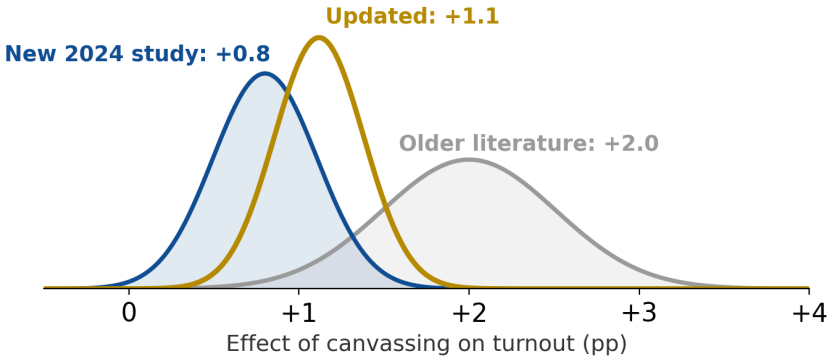
$$\text{updated SE} = \frac{1}{\sqrt{14.51}} = 0.26$$

Then build the interval the usual way, estimate ± 1.96 SE:

$$\text{updated 95\% CI} = +1.1 \pm 1.96 \times 0.26 = +1.1 \pm 0.5 = [+0.6, +1.6]$$

The updated SE (0.26) is smaller than either one alone (0.51 and 0.31). More evidence, more sure.

The Updated Belief



$+1.1$ pp, interval $[+0.6, +1.6]$

What the Update Costs the Donor

What you believe	Effect	Cost per vote	Votes from $500K
Older literature only	+2.0 pp	$6/0.020 = $300	1,667
Updated	+1.1 pp	$6/0.011 = $545	917
New study only	+0.8 pp	$6/0.008 = $750	667

Should She Put It All in Mail?

Mail is \$86 a vote. \$500K buys 833,333 mailers, and $833{,}333 \times 0.007 = 5{,}833$ votes.

That assumes every mailer works as well as the first one.

Assume her races have 200,000 households worth mailing. 833,333 pieces is about four to every household.

Does the fourth mailer to the same house do what the first one did?

Diminishing Returns

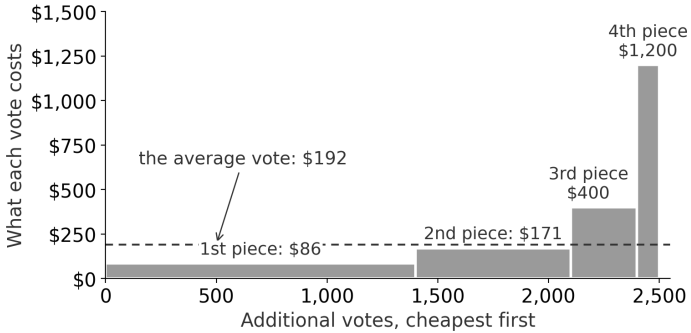
Diminishing returns: each additional dollar buys less than the dollar before it.

The $+0.7$ pp was measured on people who got *one* mailer. Assume each piece does less than the one before:

Piece	Effect (assumed)	Cost	Votes	Cost per vote
1st	$+0.70$ pp	\$120,000	1,400	\$86
2nd	$+0.35$ pp	\$120,000	700	\$171
3rd	$+0.15$ pp	\$120,000	300	\$400
4th	$+0.05$ pp	\$120,000	100	\$1,200

Each round costs $200{,}000 \times \$0.60 = \$120{,}000$. Votes are $200{,}000\times$ the effect: $200{,}000 \times 0.007 = 1{,}400$.

The Average Vote and the Last Vote



Four pieces each: \$480,000 buys $1{,}400 + 700 + 300 + 100 = 2{,}500$ votes. The 5,833 priced every piece like the first.

- **The average vote:** total cost over total votes. $\$480{,}000/2{,}500 = \$192$.
- **The last vote:** what the last round costs per vote. Here, \$1,200.

Price the Last Vote

From this week's reading

Lackey & Singer, "Why the Cost-Per-Vote Formula Needs to Change": what matters is "how much the last vote costs," not "the average cost of obtaining a single vote." Buy the cheapest votes first, but stopping there "will rarely secure a win. It's more likely to get to 48 percent of the vote in a two-way race when 50-percent plus one is needed."

For the donor: keep mailing while the next mailer is the cheapest vote she can buy. A canvassing vote costs \$545. The third mailer (\$400 a vote) beats it; the fourth (\$1,200) does not. So: three mailers, then spend what is left on doors.

Not Every Vote Is Worth the Same

A vote only matters if it could change who wins.

Moving a candidate from 89% to 90% "has no impact on the outcome" (Lackey & Singer).

So the donor should spend in races that could actually be close. More in Week 12!

Takeaway

- The best tactic *per person* is not the best tactic *per dollar*. Cost per vote needs both the cost and the effect.
- Cost per vote = cost / effect.
- An additional voter is not a net vote. At 75–25 it takes two of them, so targeting matters (next few weeks).
- Cost per vote has a confidence interval too.
- Bayesian updating is a weighted average, and the weight is precision, $1/\text{SE}^2$. You move toward whichever belief is narrower.
- Cheap votes run out. The last vote costs more than the average one, and that is why the answer is usually a mix.

Today and Your Final Project

Your budget table has a column called "What it buys," and the answer is net votes in a named state. Today was how you get that number.

- **Cost per vote:** price each line item per attempt, with an effect measured on the same people.
- **Net votes:** an additional voter is not a net vote. Persuasion counts twice.
- **Uncertainty:** price the interval, not just the point estimate. When studies disagree, weight them by precision. Which one would you bet on?
- **The last vote:** $250 million is a lot of money. What does the *last* dollar in each line buy, not the average one?
- **Where:** a vote in a state that will not decide the election is not worth buying.

Lying With Data This Week

Lying with data: Ranking tactics by effect size or cost per attempt/delivery and not by cost per vote.

Read before next week

Next week: turnout models, and how to grade one. Pew Research Center, "Q&A: The growing use of 'voter files' in studying the U.S. electorate." Pew, *Commercial Voter Files and the Study of U.S. Politics*, the chapter on demographic data: how accurate a vendor's modeled variables actually are. And Google's Machine Learning Crash Course, "Classification: Accuracy, recall, precision, and related metrics."

The Checklist So Far

1. **W1:** Conflating fixed and marginal costs.
2. **W2:** Presenting an observational comparison as a causal effect.
3. **W3:** Applying a result from one setting to a different one.
4. **W4:** Cherry-picking the winning arm from a multi-arm test.
5. **W5:** Treating an underpowered null as evidence of no effect.
6. **W7:** Reporting the complier comparison as a causal effect.
7. **W8:** Cherry-picking polls; house effects; ignoring nonresponse bias.
8. **W9:** Ranking tactics by effect size or cost per attempt instead of cost per vote.

Break: 5 Minutes

$\sim$ 5 min $\sim$

Live Coding

Open `livecode.ipynb`.

Five tasks:

1. Load the tactics data and compute cost per vote
2. Fix the units on canvassing and phone; watch $1,000 become $300
3. Compute total votes produced per $500K
4. Show how uncertainty changes the ranking
5. Do the Bayesian update and re-price canvassing

Problem Set This Week

Open `problem_set.ipynb`.

1. Compute cost per vote for all four tactics (pre-filled).
2. **Write the cost-per-vote table yourself** for all four tactics plus the new study.
3. **Do the Bayesian update yourself.** Report the updated belief and the weights.
4. Compute total votes under two scenarios: (a) all-canvassing at the new study's numbers, (b) all-mail.
5. Write a **memo to the donor** (250–350 words): how should she allocate $500K?